

Phase 1 — Data Acquisition

How the data is captured and its sources

The project draws on two datasets (Table 1). ORNL Summit power data — the **demand** side — is captured **passively** by on-node hardware sensors — NVIDIA DCGM for GPU power and the Baseboard Management Controller for power-supply draw — and published as Apache Parquet. TVA grid generation — the **supply** side — is pulled **actively** from the U.S. EIA API.

Source	Side	What it provides	How captured	Format / access
ORNL Summit	Demand	Per-node power · 4,626 nodes	Hardware sensors (DCGM + BMC)	Apache Parquet
TVA grid (via EIA)	Supply	Hourly generation by fuel type	FERC Order 830 reporting	EIA REST API (JSON)

Table 1 — The two data sources, stated explicitly.

The two datasets in brief

Dataset	Granularity	Key columns used
ORNL Summit (Parquet — 12 of 71 cols)	per node, per minute	timestamp · hostname · node_state · 6× GPU power (p0_gpu0...p1_gpu2_power) · 2× PSU i
TVA via EIA (long format)	per fuel, per hour	period · fueltype (COL, NG, NUC, OIL, SUN, WAT, WND, OTH) · value (MWh) · value-units

Table 2 — The fields actually used from each dataset.

Evaluating source reliability

Both primary sources are highly authoritative — a U.S. national laboratory (peer-reviewed) and a federal agency reporting under legal mandate — and come from direct measurement rather than estimates, with completeness of 99.98% (ORNL) and 99.6% (EIA), cross-validated against the source paper and TVA's own reports.

Options considered and chosen approach

For demand, synthetic, NERSC and cloud data were rejected as unrepresentative, unavailable at resolution, or not public at node level; ORNL alone offered peer-reviewed per-node, per-minute power. For supply, paid APIs and other grids were rejected on cost, reproducibility and geography, so the free official EIA API was chosen (Table 3). A **data-first discipline** is maintained: raw data is never modified, every load is validated, and anomalies are logged, not dropped.

Data	Options rejected	Why rejected	Chosen
Demand	Synthetic / NERSC / cloud	Unrepresentative · unavailable at resolution · not public at node level	ORNL Summit
Supply	Electricity Maps / WattTime / other grids	Grids not reproducible · wrong geography	EIA API (TVA)

Table 3 — Alternatives weighed against the chosen sources.